

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	Shri Shanmuga Priya K
Register Number	192324035

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I Shri Shanmuga Priya K [192324035] certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student :

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

- Type-1 (Bare Metal): Installs directly onto physical hardware, bypassing a host OS entirely.
- Performance: Delivers superior performance with near-native hardware speeds and minimal latency because virtual machines communicate directly with physical components.
- Security: Offers a significantly reduced attack surface. The absence of a host OS means fewer vulnerabilities, enforcing strict isolation between the hardware and guest environments.
- Manageability: Built for enterprise scale, allowing centralized orchestration, automated clustering, and live migrations.
- Type-2 (Hosted): Operates as an application on a traditional host OS like Windows or Linux.
- Limitations: Suffers from I/O latency since resources must route through the host OS. Security is inherently weak; compromising the host OS compromises every virtual machine.
- Justification: For mission-critical cloud datacenters, Type-1 is the only viable choice. It guarantees the absolute performance, robust isolation, and high availability necessary for enterprise applications, whereas Type-2 is suitable only for consumer testing.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

- Compute Isolation: Hypervisors use hardware extensions (Intel VT-x, AMD-V) to enforce strict privilege rings. This hardware-assisted virtualization prevents guest VMs from executing privileged instructions on the shared host CPU.
- Memory Isolation: Technologies like Second Level Address Translation (SLAT) create exclusive memory boundaries. Physical memory is mapped independently to each VM, ensuring applications cannot access unauthorized memory blocks.
- Network Isolation: Software-defined virtual switches (vSwitches) and logical VLANs segregate traffic. Strict micro-segmentation rules drop unauthorized cross-department communication before it leaves the hypervisor.

- Attack Surface 1: VM Escape: A critical vulnerability where an attacker exploits flaws in the hypervisor code. This allows them to break out of their isolated guest environment and achieve root-level control over the physical host and neighboring VMs.
- Attack Surface 2: Side-Channel Attacks: An exploit targeting shared physical hardware resources. Attackers analyze shared CPU caches to infer and extract sensitive cryptographic keys or data from adjacent VMs.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

- Primary Bottleneck: The critical issue is the 35 ms storage I/O latency. Enterprise environments require latency under 10 ms. This indicates severe storage contention and starvation.
- Boot Storm Impact: The excessive 170-second VM boot time directly results from this storage latency. Simultaneous read/write requests from multiple VMs are overwhelming the underlying storage controller.
- CPU Masking: The 92% CPU utilization is artificially inflated. Processors are likely stuck in high "I/O Wait" states, burning idle cycles waiting for the mechanical disk or overloaded array to return requested data.
- Action 1 - Storage Migration: Immediately migrate mission-critical or high-IOPS virtual machines to an all-flash NVMe/SSD storage tier to relieve the existing mechanical disk bottleneck.
- Action 2 - Live Migration: Execute live migrations (vMotion) to move running VMs to underutilized hosts, dropping CPU and memory utilization below the safe 80% threshold.
- Action 3 - Storage QoS: Implement strict Storage Quality of Service limits to throttle non-critical VMs.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

- **Traditional Limitations:** Traditional datacenter architectures rely on rigid physical hardware components. Scaling requires manual server racking, physical cabling, and localized device configuration, which drastically limits operational speed.
- **SDDC Core Architecture:** An SDDC abstracts all physical hardware infrastructure into programmable software. This shift enables instantaneous, API-driven provisioning and automated resource scaling without physical intervention.
- **Compute Agility:** Physical servers are pooled into logical resource clusters. Virtual machines and containers can be deployed, scaled, or terminated in seconds to match application demand dynamically.
- **Storage Agility (SDS):** Software-Defined Storage decouples management logic from physical drives. Administrators apply dynamic storage policies and automated tiering across heterogeneous arrays without manually reconfiguring complex hardware RAID setups.
- **Network Agility (SDN):** Software-Defined Networking replaces physical routers and firewalls with logical constructs. Entire subnets, load balancers, and complex micro-segmentation security policies deploy instantly alongside workloads, ensuring security scales automatically with the application, ensuring immediate compliance.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

- **Root Cause:** The multi-cloud federation project failed because disparate cloud providers utilize incompatible directory schemas and proprietary access protocols, preventing seamless cross-platform user authentication and role mapping.
- **Federated Trust Broker:** To resolve this, architects must deploy a centralized identity broker using standardized federation protocols like SAML 2.0 or OpenID Connect (OIDC).

- Token-Based Exchange: Instead of transmitting vulnerable raw user passwords and credentials across network boundaries, the Identity Provider (IdP) and Service Provider (SP) exchange cryptographically signed authorization tokens.
- Claims-Based Privacy: To maintain stringent data privacy, the architecture uses claims-based authorization. The IdP transmits only essential attributes (claims) required for access, shielding the full organizational user database from third-party cloud environments.
- Just-In-Time Provisioning: The Service Provider dynamically generates temporary, least-privilege user profiles upon successful authentication. These profiles are automatically destroyed post-session, eliminating the security risk of redundant, persistent accounts across multiple cloud platforms.